



Cambridge IGCSE™

DESIGN AND TECHNOLOGY

0445/41

Paper 4 Systems and Control

May/June 2026

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **15** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

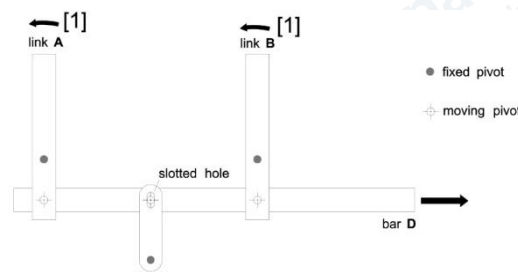
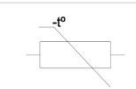
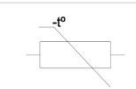
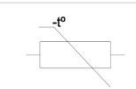
The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	Unclear
	Benefit of the doubt
	Incorrect point
	Error carried forward
N/A	Highlighting areas of text
	No benefit of doubt given
N/A	Off-page comment – allows comments to be entered off the page
	Repeat
	Indicates that the point has been noted, but no credit has been given
	Indicates that the point has been noted, but no credit has been given (big)

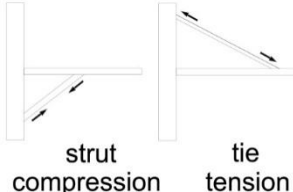
Annotation	Meaning
	Correct point
	Too vague
	Relevant detail

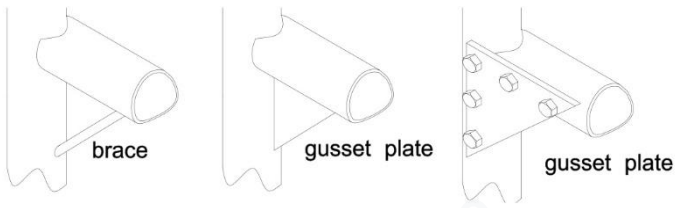
Question	Answer	Marks	Guidance
Section A			
1(a)	Any two benefits [2 × 1] Benefits of using plastics could include: • Consistent quality of material • Durable / maintenance free • Dimensional stability in wet conditions • Lighter than hardwood • Waterproof / resistant to weather damage / insect damage.	2	Allow other valid response Do not allow benefits relating to manufacture or recycling.
1(b)	Any two benefits [2 × 1] Benefits of using hardwood could include: • Material can come from a managed resource • Aesthetic qualities of hardwood • No special manufacturing facilities required • Durable if maintenance is carried out.	2	Any other valid response
2	Static load is a load that is constant / does not move / remains stationary [1]	1	Award mark for understanding shown.
3	Beam shown resting on narrow side/edge. [1] Increased number of supports / reduced distance between supports below the beam [1] Gusset plates [1] Triangulation strut between the bottom of one post and the top of the opposing post [1]	2	Any other valid response Suitable sketch [1] Explanatory notes [1] For fully justified notes award 2 marks
4	Shear force is applied to the aluminium [1]	1	
5	Any valid examples of third order levers [2 × 1]	2	Examples will include shovel, fishing rod, tweezers, tongs, pencil, chopsticks, human limbs. Any other valid response

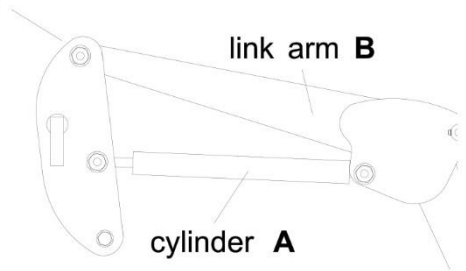
Question	Answer	Marks	Guidance												
6(a)	 Movement arrows as shown [2 × 1]	2	Allow straight arrows pointing towards left. Allow arrows to show rotation of links.												
6(b)	Linear [1] movement is converted to oscillating [1] movement	2													
6(c)	As bar D is moved to the right it will move vertically due to the oscillation of links A and B [1] Movement of bar D is limited by depth of slotted hole The slotted hole at the top of link C allows for the vertical movement [1]	2	Award marks for understanding shown.												
7	<table border="1"><thead><tr><th>symbol</th><th>component name</th><th>unit</th></tr></thead><tbody><tr><td></td><td>ammeter</td><td>ampere / amp</td></tr><tr><td></td><td>Battery</td><td>Volt</td></tr><tr><td></td><td>thermistor</td><td>ohm</td></tr></tbody></table> [4 × 1]	symbol	component name	unit		ammeter	ampere / amp		Battery	Volt		thermistor	ohm	4	Allow 'A' for Amps. Allow Ω (ohms) symbol
symbol	component name	unit													
	ammeter	ampere / amp													
	Battery	Volt													
	thermistor	ohm													
8	A toggle switch will remain in the changed position when operated [1] A push-to-make switch is momentary action [1]	2	Award marks for understanding shown.												

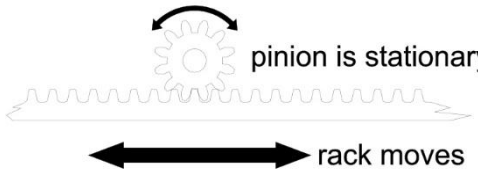
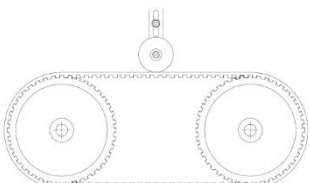
Question	Answer	Marks	Guidance
9(a)	Parallel connection [1]	1	
9(b)	Advantages of parallel connection are: If one component fails, the other will continue to work [1] With a series connection if one component fails neither will work Components can be individually controlled with their own switches [1]	2	Any other valid response Description that includes two points, 2 marks.

Question	Answer	Marks	Guidance
Section B			
10(a)(i)	The function of the counterweights is: - To reduce load on the cable drive system - Help to maintain equilibrium when the bridge deck is raised. - Reduce wear on the lifting system - Makes lifting the load more efficient.	2	[2 × 1] for 2 points mentioned. Award 2 marks for a single point fully described.
10(a)(ii)	Any one reason for using cast steel including: - Heavy - More durable than concrete - Greater precision in the weight of each piece	1	Any other valid response
10(a)(iii)	The effect of the strain on a cable needs to be tested. [1] Mention of Factor of Safety [1] Each cable needs to be capable of resisting similar strain in order to maintain an even lift. [1]	2	Any other valid response [2 × 1] for 2 points mentioned. Award 2 marks for a single fully justified point.
10(a)(iv)	Change in length = 0.01 m (10 mm) / original length = 10 m (10 000 mm) [1] $10 / 10\,000 = 0.001$ [1]	2	Award 2 marks for the correct answer with no working.

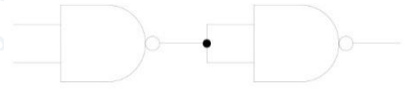
Question	Answer	Marks	Guidance
10(b)(i)	Examples of static load include: - Materials used in construction of the bridge deck - Any load placed on the bridge that does not change or remains constant - Weight of the bridge structure acting on the ground surface. Examples of moving load include: - Traffic on the bridge - Pedestrians on the bridge - Lateral forces / wind force - Rain / weight of water acting on the bridge - Earthquake. [2 × 2]	4	Any other valid response
10(b)(ii)	Explanation should include reference to: - The design load and the allowable load that can be carried by the bridge deck. - Relevant example used - Ultimate strength of bridge deck / Allowable load on bridge deck.	2	2 × 1 points included or allow 2 marks for one point fully explained.
10(c)(i)	 strut compression tie tension Accurate sketch of strut. [1] Force arrows pointing inward. [1] Force resisted is compression. [1] Accurate sketch of tie. [1] Force arrows pointing outward. [1] Force resisted is tension. [1]	6	

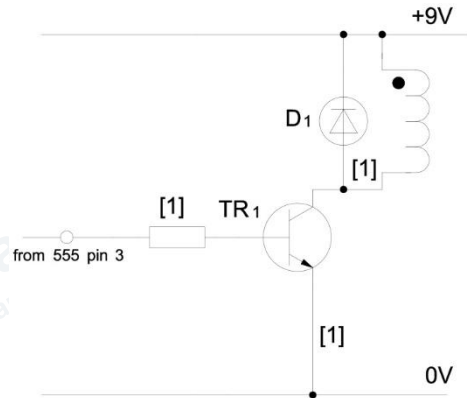
Question	Answer	Marks	Guidance
10(c)(ii)	Stages in permanently joining tubes will include: - Shaping the end of horizontal tube to fit profile of vertical tube - Cleaning area around joint - Welding or brazing the joint. 2 × 1 marks for relevant stages	2	Any other valid response
10(c)(iii)	 brace gusset plate gusset plate	2	Allow any other valid response for reinforcement method. Method of reinforcing the joint. [1] Method of fixing reinforcement into place. [1]
10(d)	Temporary methods allow: - Repositioning of parts - Easy transport before joining / assembly on site - Removal of parts after construction is completed - Easier to disassemble during the deconstruction (recycling) at the end of the structure life - Maintenance is easier / replacement of old parts. 2 × 1 marks	2	Any other valid response

Question	Answer	Marks	Guidance
11(a)(i)	fulcrum  link arm B cylinder A	1	Award mark if the fulcrum is marked on Fig 11.1.
11(a)(ii)	Description may include the following: • The link arm will rotate anticlockwise around the fulcrum as the cylinder is extended. • The front loader bucket will be lifted/raised. • Cylinder A will be resisting compression 2 × 1 marks for valid points	2	Allow 2 marks for a detailed description
11(a)(iii)	Benefits of using a grease nipple will include: • Ease of access to a grease gun • Easy connection that prevents wastage of grease • Keeps the component interior clean • Allows grease to be applied under pressure	1	Award 1 mark for any relevant benefit. Any other valid response
11(a)(iv)	The following may be included in the explanation: • There is a relatively small amount of movement • Only a small amount of heat is produced • Plain bearing is more durable • The joint is greased • More economical than other types of bearing • Plain bearings can support a heavy load moving at low speed.	2	Award marks for any two relevant points, two marks for a single point fully explained.

Question	Answer	Marks	Guidance
11(a)(v)	 pinion is stationary rack moves The pinion rotates in either direction as steering wheel is turned. [1] The rack moves side to side [1] converting rotary motion to linear motion. [1]	3	
11(b)(i)	Clockwise arrow on the 40t output gear.	1	
11(b)(ii)	Reduction from worm gear to 40t gear = 40:1 [1] Reduction from 15t gear to 40t gear = 2.67:1 [1] Reduction from 25t gear to 40t gear = 1.6:1 [1] Total reduction = $40 \times 2.67 \times 1.6 =$ 170.89:1 or 170.67:1 [1]	4	Accept numerical answer not expressed as a ratio.
11(b)(iii)	Speed of output gear = $600 / 170.89$ [1] = 3.51 rpm or 3.52 [1]	2	Allow ecf from (b)(ii)
11(b)(iv)	Loss of efficiency could be caused by: Friction, heat loss or sound. 2 × 1 marks	2	
11(c)(i)	 Jockey wheel to put pressure on the belt (above or below) [1] Pressure placed on outside of belt [1] Method of securing jockey wheel – bolt in slot, spring to apply pressure [1]	3	Any other valid response Do not allow method that involves moving either pulley. Do not allow pressure on the inside of belt.

Question	Answer	Marks	Guidance
11(c)(ii)	Advantages of a toothed belt include: • No slipping • Delivers more power • Pulleys kept at a constant alignment with each other • Longer life than vee belt. • Consistent torque at output [2 × 1]	2	Any other valid response
11(c)(iii)	Advantages of a sprocket and chain include • Longer life / more durable than a belt • Can operate in poor environment • Less affected by heat • Both sides of the chain can be used. [2 × 1]	2	Any other valid response
12(a)(i)	Description to include any two of the following points. • 14 pin • Dual-in-line • Quad gates • NAND gate.	2	1 mark for each valid point made [2 × 1]
12(a)(ii)	Pin 14 is the positive power connection [1]	1	
12(a)(iii)	Identification of pin 1 on a logic IC can be: • Notch on the end of the IC, then first pin on the left is pin 1 • Small dot on the IC body next to the pin. [2 × 1]	2	

Question	Answer	Marks	Guidance
12(a)(iv)	NAND gates used [1] NAND followed by NOT. [1] 	2	
12(b)(i)	Sequence 1 5 4 3 7 6 2 8	3	1 mark for each pair in correct sequence up to 3 marks.
12(b)(ii)	Description to include: - Heat the joint with soldering iron or desoldering tool - Use 'solder sucker' / braid or similar to remove excess solder - Push resistor legs out from solder side or lever out from component side - Remove any excess solder from holes.	3	1 mark for each point mentioned in description to a maximum of 3.
12(b)(iii)	Other methods of connection include: - Plug and socket - Screw terminal - PCB plug / boxed header / IDC connectors	1	Any other valid response
12(c)(i)	A monostable will provide a single output pulse when triggered, it has only a single stable state .	1	Award mark for understanding shown.

Question	Answer	Marks	Guidance
12(c)(ii)	Use of the formula $t = 1.1 \times R \times C$, $t = 8 \text{ s}$, $C = 100 \mu\text{F}$ [1] $R = 8 / 1.1 \times 100 \mu\text{F}$ [1] $R = 72\,727 \Omega$ or $72.73 \text{ k}\Omega$ [1]	3	Award 3 marks for the correct answer with no working.
12(c)(iii)	The tolerance in the capacitor can be overcome by using a variable resistor / potentiometer in place of R_2 in the circuit [1] Value of VR would be $100 \text{ k}\Omega$. [1]	2	
12(c)(iv)		3	
12(c)(v)	Diode is included to prevent damage to the transistor [1] caused by back emf from the solenoid coil. [1]	2	Award marks for understanding shown.